DressMe BO6

Smart Wardrobe Digitization - summary of the executed notebook and its results

Objective. Develop two ML components for a digital wardrobe: classify a garment from one photo and produce a short automatic description. The notebook audits the dataset, trains and compares classifiers, adds attribute predictions, verifies the export and implements optional photo-quality warnings.

1. Dataset audit and preparation

The Fashion Product Images dataset was used. Duplicate dataset copies were identified, and one canonical copy was selected. Metadata, per-product records and images were inspected before constructing the modelling pool.

Audit item	Recorded result
Metadata	44,446 records: 44,424 well-formed rows and 22 malformed rows repaired.
Image integrity	44,441 images fully decoded; no decoding failures.
Image / metadata matching	5 metadata records without images; no orphan images.
Exact duplicates	652 groups across the dataset; 5 mapped rows with conflicting targets excluded.
Final modelling pool	15,441 mapped garment rows with decoded images.

Taxonomy: dress, jacket, jeans, shirt, shorts, skirt, sweater, sweatshirt, t-shirt, top and trousers. Catalog Track Pants were mapped to trousers; unsupported source categories were excluded.

2. Split and preprocessing

Split	Rows	Share
Training	10,795	69.9%
Validation	2,310	15.0%
Reserved test	2,336	15.1%

Identical SHA-256 image copies were grouped into the same split: **zero exact-image groups cross splits**. Training contained 4,931 t-shirts and 88 skirts, so moderate class weights were applied.

Images were EXIF-corrected, converted to RGB, padded to a square, resized to **224 x 224** and normalised with ImageNet statistics. Training used mild flips, rotation and colour jitter. A pixel cache avoided repeated decoding and resizing.

Training ran on a Kaggle Tesla T4. The reserved test split remained closed; the model comparisons below use validation data. Near-duplicate/product leakage remains an unresolved limitation.

Classification experiments

Transfer learning, model comparison, limited tuning and selection

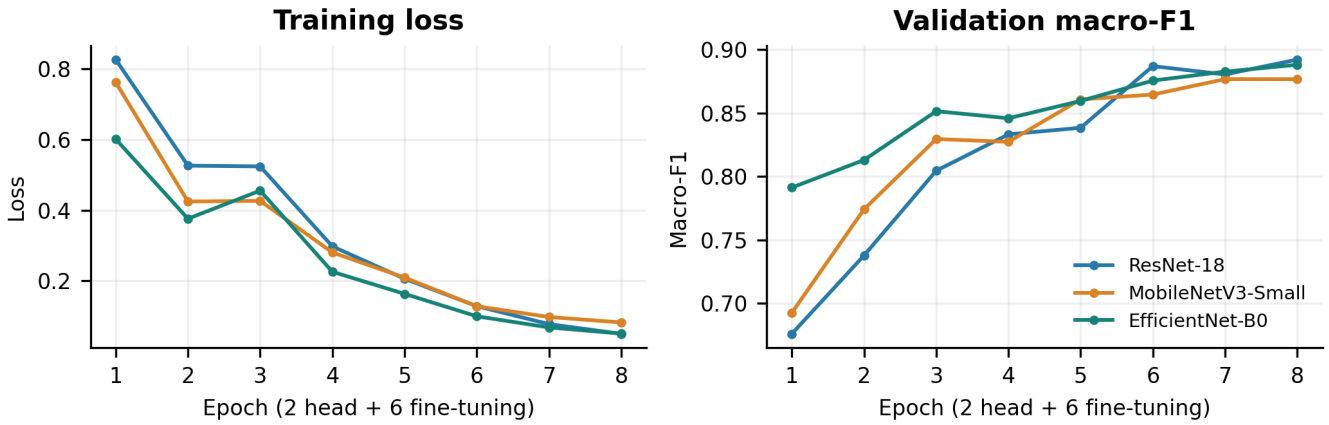
3. Training approach

Three ImageNet-pretrained models were compared using the same data and preprocessing. Each main experiment trained the new classification head for **2 epochs** with the backbone frozen, then fine-tuned the final blocks for up to **6 epochs**, using batches of 32. Best validation checkpoints were retained.

Model	Macro-F1	Accuracy	Top-2	Weights	CPU*
Majority: t-shirt	0.0572	45.93%	-	-	-
ResNet-18	0.8923	92.94%	98.87%	44.81 MB	30.64 ms
MobileNetV3-Small	0.8768	91.43%	98.61%	6.25 MB	10.94 ms
EfficientNet-B0	0.8882	93.16%	99.22%	16.39 MB	28.21 ms

All scores are measured on 2,310 validation images. Macro-F1 averages the F1 score of each class equally, so rare classes contribute as much as frequent ones. *CPU timings cover only the classifier forward pass on the Kaggle CPU.

Selected model: EfficientNet-B0. The selection rule chose the fastest CPU candidate within 0.01 macro-F1 of the best. EfficientNet reached **93.16% accuracy, 0.8882 macro-F1 and 99.22% top-2 accuracy**, with smaller weights than ResNet-18.



Recorded training loss and validation macro-F1 across the eight epochs of the main experiments.

4. Additional experiments and error analysis

Two ResNet tuning trials were completed: longer fine-tuning obtained **0.8882 macro-F1**, and doubling the backbone learning rate obtained **0.8858**. Neither improved on the original 0.8923 result. A repeat with seed 43 reached **0.8757**, showing variation between runs. A third tuning trial was skipped.

The selected model made **158 validation errors**. T-shirt/top confusion accounted for 73. Lower F1 scores included sweatshirt (0.771), sweater (0.800) and top (0.802). Skirt F1 was 0.833, based on only 21 validation examples. Per-class metrics, confusion plots and error-image grids were saved.

Automatic descriptions

Attribute predictions, structured fields and deterministic text rendering

5. Description pipeline

Three small linear heads were trained on training-row catalog labels to predict **sleeve length, pattern and colour**. They reuse the selected classifier's features plus a colour histogram, normalised using training statistics. Missing and unsupported attribute labels were excluded from the relevant head evaluation.

Descriptions are generated by a **deterministic template**, rather than a generative language model. It renders supported structured fields and omits unknown values. Colour remains a suggestion until the user confirms it. Every prediction is flagged for user review, and corrections can override model fields.

Attribute	Validation result	Description policy
Sleeve length	92.74% agreement; macro-F1 0.8718; 1,639 usable labelled rows.	Stated for supported categories: jacket, shirt, sweatshirt, t-shirt and top.
Pattern	82.67% overall agreement on 1,835 labelled rows. At threshold 0.97: 72.53% coverage and 9.32% error.	Accepted above score 0.97 for supported categories. The word solid is omitted from the sentence.
Colour	69.05% agreement; macro-F1 0.6427; 2,291 usable labelled rows.	Two suggestions offered; enters the sentence after user confirmation.

These metrics measure agreement with weak catalog metadata, not human-verified factual correctness. Threshold exploration reused validation data.

6. Generated description results

Sentence outcome on all validation images	Count
Category + two attributes mentioned	998
Category + one attribute mentioned	831
Known category only	375
Category withheld: A garment.	106
Total	2,310

1,829 descriptions (79.2%) mention at least one attribute beyond category. Accepted structured values and words actually rendered are counted separately; for example, a solid pattern can be present in the fields without appearing in the sentence.

Examples recorded in the outputs:
Product 10065: A printed short-sleeved t-shirt.
Product 16392: A short-sleeved shirt.
Product 39386: A pair of jeans.

Quality checks and export

Optional image-quality assistant and the final notebook outputs

7. Optional image-quality assistant

A rule-based module measures resolution, sharpness, brightness and contrast, and returns warnings with suggested actions. It requires no additional trained model. Readable-image warnings are advisory; unreadable inputs are reported separately.

The blur threshold was calibrated on **300 validation photos** and blurred copies. Evaluation used **60 of those photos**, each with four strong synthetic degradations. It therefore demonstrates rule behaviour on calibration-related examples, rather than independent phone-photo performance. The chosen sharpness threshold was **13.84**.

Image set	Images	Observed result
Unmodified catalog photos	60	0% returned any warning.
Strongly blurred	60	100% returned the expected blur warning.
Strongly darkened	60	100% returned the expected darkness warning.
Strongly brightened	60	100% returned the expected brightness warning.
Downscaled	60	100% returned the expected resolution warning.



Saved notebook example: an unmodified photo and its blurred, darkened, brightened and downscaled variants.

8. Exported artifacts and verification

selected_model_bundle.zip: 25 files, approximately 15.3 MB compressed. Includes classifier weights, attribute heads, inference and description code, class mapping, preprocessing, review policy, quality checks and checksums.
bo6_results.zip: approximately 238 MB, containing experiments, metrics, figures, candidate model exports and the selected bundle.

The core export passed **22/22 software checks** in a fresh process using 3 synthetic and 44 validation images. After adding quality checks, the portable model self-test passed again. These checks establish software consistency; real-world performance remains to be evaluated.

Remaining evaluation: reserved catalog test, independent DressMe phone photos and human review of description factuality. Near-duplicate/product leakage remains unresolved. Quality thresholds are provisional. The notebook delivers both mandatory ML components and an optional quality assistant; application integration is the next phase.